Team Selection Test Semester 2

Paly Math Club

2025-26

This test is the team-selection test for **Stanford Math Tournament** and **Carnegie Mellon Informatics and Mathematics Competition**.

The following are brief information of the competition:

SMT	CMIMC
April 18, 2026	March 28, 2026
Cost: $0	Cost: $800-1000
Location: Stanford	Location: CMU

Your decision on SMT and CMIMC is BINDING. Withdrawing on short notice may prohibit you from participating in future tournaments.

Please circle the competitions you are planning to attend:

1) SMT 2) CMIMC

Instructions: This is a **60** minute test with **15** questions. Each question is given 1 point. No point deduction for incorrect answers, no partial credit except the estimation problem.

Friendly Reminder: The perceived difficulty of problems varies from person to person. If you find one problem challenging, move on and try the others — you may find them more accessible.

Name: _____ Grade: _____

Please list any of your relatives that might be able to chaperone for competitions, including which competitions they would be available for and their relation to you (ex. father). (This potentially may be used in team selection if we are severely lacking chaperones.)

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1 Algebra

1. Let a and b be positive integers such that $a + ab = 1443$ and $ab + b = 1444$. Find $10a + b$.
2. Consider the recursive sequence defined by $a_{n+1} = (a_n)^n + 1$, with $a_1 = 0$. What is the last digit of a_{2026}?
3. Let z be a complex number satisfying $(z + \frac{1}{z})(z + \frac{1}{z} + 1) = 1$. Evaluate $(3z^{100} + \frac{2}{z^{100}} + 1)(z^{100} + \frac{2}{z^{100}} + 3)$.
4. Find all ordered triples (a, b, c) of positive reals that satisfy

$$\lfloor a \rfloor bc = 3, \quad a\lfloor b \rfloor c = 4, \quad ab\lfloor c \rfloor = 5,$$

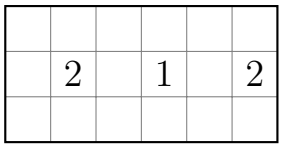
where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x.

2 Geometry

1. What is the distance from the centroid to the incenter of a $3 - 4 - 5$ right triangle?
2. You have two rhombii, $ABCD$ and $EFGH$, such that $\angle ABC = 60^\circ$ and $\angle EFG = 30^\circ$. In addition, you can inscribe a circle of radius 1 inside each rhombus. The product of the area of $ABCD$ and the area of $EFGH$ can be expressed as $\frac{a}{\sqrt{b}}$ where a and b are positive integers, and b is squarefree. Find $a + b$.
3. Let ABC be a triangle with $AB = 16$ and $AC = 5$. Suppose the bisectors of angles $\angle ABC$ and $\angle BCA$ meet at point P in the triangle's interior. Given that $AP = 4$, compute BC.
4. Let $ABCDE$ be a square pyramid of height $\frac{1}{2}$ with square base $ABCD$ of side length $AB = 12$ (so E is the vertex of the pyramid, and the foot of the altitude from E to $ABCD$ is the center of square $ABCD$). The faces ADE and CDE meet at an acute angle of measure α (so that $0^\circ < \alpha < 90^\circ$). Find $\tan \alpha$.

3 Combinatorics

1. Inhoo is playing the game of Minesweeper. A number on a square denotes the number of mines that share at least one vertex with that square. A square with a number may not have a mine, and the blank squares are undetermined. How many ways can the mines be placed in this configuration?



2. Aiden is creating a collection of sets such that each set's elements are integers between 1 and 28 inclusive. Every pair of 2 sets in the collection must have a non-empty intersection, and every triplet of 3 sets in the collection must have an empty intersection. What is the maximum number of sets in Aiden's collection?
3. Values $a_1, \ldots, a_{2013}$ are chosen independently and at random from the set $\{1, \ldots, 2013\}$. What is the expected number of distinct values in the set $\{a_1, \ldots, a_{2013}\}$? You may leave your answer in exponential and fractional form.

4 Number Theory

1. Suppose a and b are the legs of a right triangle with a hypotenuse of length 257. If a and b are integers, what is the sum of the minimum and maximum values of a?

2. Find the number of positive integer divisors of 12! that leave a remainder of 1 when divided by 3.

3. Sky is solving a math problem, and he encounters the expression $\sqrt{15!}$. He attempts to simplify this radical by expressing it as $a\sqrt{b}$, where a and b are positive integers. The sum of all possible distinct values of ab can be expressed in the form $q \cdot 15!$ for some rational number q. Find q.

5 Estimation

Four lines are chosen independently as follows: for each line, two points are chosen independently and uniformly at random on the circumference of a unit circle, and the unique line through those two points is taken (the line is extended to be infinitely long). Let p be the probability that no two of the six lines intersect at any point whose distance from the center of the circle is at most 2.6. Find the least positive integer N such that $Np > 1$. [1]

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[1] Your score is scaled to the interval $[0, 1]$ by

$$S = \max\left(1 - \pi \left|\ln\left(\frac{A}{d}\right)\right|,\ 0\right),$$

where A is the value you submit and d is the value that my code gave me.